APOM-DAPP — Project Description

Decentralized Gaming & DeFi Platform — Full Scope & Vision

*This document describes APOM-DAPP as a **large-scale, multi-year platform initiative**. It is intended for candidates, partners, and stakeholders to understand scope, architecture, and what "building this properly" entails.*

Part 1 — Vision & Strategic Context

1.1 What Is APOM-DAPP?

APOM-DAPP is a **decentralized gaming and DeFi platform** that aims to become a **unified Web3 destination** where users can:

- **Play** — Engage in play-to-earn games, own in-game assets as NFTs, compete in tournaments, and earn the native APOM token.
- **Trade** — Swap tokens, provide liquidity, stake, and farm yield across multiple chains and pools.
- **Collect** — Buy, sell, and trade NFTs (gaming items, art, virtual land) in a cross-chain marketplace.
- **Invest** — Participate in vetted IDOs and early access to new gaming and DeFi projects via the launchpad.
- **Govern** — Shape the platform and tokenomics through transparent, on-chain or hybrid governance.

The platform is **multi-chain by design**: it does not tie users to a single network but supports Ethereum, Polygon, BSC, Arbitrum, Avalanche, Fantom, and other EVM-compatible chains, with a consistent experience and shared identity where possible.

1.2 Why This Is a Long-Term Project (1+ Year)

APOM-DAPP is **not a small DApp or MVP**. It is conceived as a **platform** with:

- **Five major product pillars** (Gaming, DeFi, NFT Marketplace, Launchpad, Governance), each with its own feature set, integrations, and lifecycle.
- **Multi-chain deployment** — Smart contracts, indexers, and APIs must work across many networks; each chain adds deployment, testing, and maintenance overhead.

- **High standards** — Security (audits, bug bounties), compliance (KYC/AML where required, jurisdictional considerations), performance (sub-second reads, reliable RPC failover), and UX (wallet onboarding, gas abstraction, mobile).
- **Ecosystem dependencies** — Integrations with wallets, oracles, bridges, game studios, and third-party DeFi protocols; each partnership or integration takes time to design, implement, and maintain.
- **Operational readiness** — Monitoring, alerting, incident response, documentation, and support processes that scale with usage and value locked.

Realistically, **reaching a mature, production-grade platform** with all pillars live, audited, and operable across multiple chains is a **12–18+ month journey**, with ongoing iteration beyond that.

1.3 Target Users & Success Vision

- **Gamers** — Play-to-earn players, guilds, and streamers who want verifiable ownership of assets and earnings.
- **DeFi users** — Traders and LPs who want gaming-themed pools, APOM-denominated incentives, and multi-chain access.
- **Collectors & creators** — NFT buyers/sellers and artists/game studios listing on the marketplace.
- **Projects** — Teams launching tokens or NFTs via the launchpad and seeking community and liquidity.
- **Token holders** — APOM holders participating in governance and platform upside.

Success means: a secure, scalable, multi-chain platform with active usage in gaming and DeFi, a thriving NFT marketplace, successful launchpad rounds, and a governed, sustainable token economy.

Part 2 — Product Pillars (Detailed Scope)

Each pillar is a **substantial product area** with multiple sub-features and its own roadmap.

2.1 Gaming Hub

Goal: Become a go-to destination for on-chain gaming and play-to-earn.

Planned scope (over time):

- **Game directory & discovery** — Curated list of integrated games (RPG, racing, strategy, simulation, etc.) with metadata, rewards (e.g. APOM/day), player counts, and status (Live / Coming Soon).

- **Unified player identity & progress** — Link wallet(s) and optional email/social; persist achievements, levels, and cross-game reputation (on-chain and/or off-chain with attestations).
- **In-game asset ownership** — NFTs for items, characters, land, vehicles; view and manage inventory in the platform; use items across supported games where applicable.
- **Rewards & economy** — Earning APOM and other tokens from gameplay; claim flows, vesting, and anti-cheat/anti-abuse mechanisms.
- **Tournaments & leaderboards** — Seasonal and one-off competitions, rankings, and prize distribution (on-chain or hybrid).
- **Game developer SDK / docs** — So third-party studios can integrate with APOM-DAPP (auth, assets, rewards) and list on the hub.
- **Social & engagement** — Guilds, friends, chat (if in scope), and notifications for rewards and events.

Why it takes time: Game integrations are per-studio; each game has different contracts and mechanics. Building a generic SDK, reward engine, and fraud controls is non-trivial and evolves with each new title.

2.2 DeFi Exchange

Goal: Full-featured DeFi experience tailored to the gaming ecosystem and APOM token.

Planned scope:

- **Token swap (AMM)** — Multi-hop swaps, slippage and deadline controls, routing across multiple pools and (where applicable) aggregators.
- **Liquidity pools** — Create/add/remove liquidity; pool types (e.g. Gaming Pool, Stable Pool, Blue Chip, New Launch); fee tiers and gauge-style incentives.
- **Yield farming & gauges** — Stake LP tokens or single-sided assets; earn APOM and/or other rewards; lock periods and boost mechanics if applicable.
- **Staking** — Stake APOM (and possibly other tokens) for governance rights and/or yield; lock-up options and early exit penalties if needed.
- **Analytics & dashboards** — TVL, APY, 24h volume, PnL, impermanent loss estimates; historical charts and export.
- **Multi-chain support** — Same UX across chains; bridge hints or integration; chain-specific contracts and RPC failover.
- **Safety & risk** — Pool due diligence, warnings, and (where possible) integration with risk oracles or audit status.

Why it takes time: DeFi is security-critical. Contract design, audits, and testing per chain are lengthy. Integrating with existing DEXs or building our own AMM and gauge system is a major undertaking.

2.3 NFT Marketplace

Goal: Primary marketplace for gaming NFTs, art, and virtual assets, with cross-chain and royalty support.

Planned scope:

- **Collections & discovery** — Browse by category (Gaming, Art, Virtual Real Estate, etc.); floor price, volume, item count; featured and trending.
- **Listing & trading** — List NFTs (fixed price, auction, bundle); buy, sell, transfer; offer/counter-offer flows if applicable.
- **Royalties & creator fees** — Enforce on-chain or marketplace-level royalties; payouts to creators and platform.
- **Cross-chain** — View and trade NFTs across chains (via bridges or multi-chain UI); unified “portfolio” view.
- **Gaming integration** — Items listed from Gaming Hub appear in marketplace; “use in game” or “list for sale” from inventory.
- **Trust & safety** — Verification badges, reported content handling, and optional curation.
- **Analytics** — Floor history, volume, top collections, and user-facing stats.

Why it takes time: NFT standards (ERC-721, ERC-1155, chain-specific), royalty handling, and cross-chain indexing/trading each require design, contracts, and backend work.

2.4 Launchpad

Goal: Vetted, fair IDO and early-access platform for gaming and DeFi projects.

Planned scope:

- **Project submission & vetting** — Application flow, due diligence (team, tokenomics, contracts), and approval workflow.
- **Campaign types** — Fixed-price sale, Dutch auction, bonding curve, or hybrid; whitelist, public, and tiered access (e.g. by APOM staked).
- **Token distribution** — Claim flow, vesting schedules, and refund logic for failed or cancelled rounds.
- **Community voting** — Optional “community choice” or signal votes before or during sales.
- **Compliance** — KYC/AML where required by jurisdiction; terms, disclaimers, and investor eligibility.
- **Post-launch** — Listing on DeFi exchange, liquidity bootstrap, and optional lock/vesting dashboards.

Why it takes time: Legal and compliance vary by region. Smart contracts for sales and vesting must be simple, secure, and audited. Vetting and operations are ongoing.

2.5 Governance

Goal: Transparent, inclusive governance for protocol and token decisions.

Planned scope:

- **Proposal lifecycle** — Create, discuss, vote, execute; quorum and voting period; optional delegation.
- **Proposal types** — Parameter changes (e.g. fees, rewards), treasury spend, grant programs, protocol upgrades, or signaling.
- **Voting power** — Typically from staked APOM (and possibly time locks); snapshot vs on-chain voting depending on cost and finality.
- **Execution** — Where applicable, timelock and multisig for execution; visibility of pending and executed actions.
- **Transparency** — Public history of proposals and votes; easy-to-read summaries and links to on-chain data.

Why it takes time: Governance design (tokenomics, quorum, delegation) and safe execution paths (multisig, timelock) need careful design and often audits.

Part 3 — System Architecture (High Level)

The system is split into **frontend**, **backend**, **smart contracts**, and **infrastructure/data** — each with long-term scalability in mind.

3.1 Frontend (Web Application)

Role: Primary user interface for all pillars; must be fast, accessible, and wallet-ready.

Current / baseline stack:

- React 18, TypeScript, Vite, React Router v6.
- Tailwind CSS, Radix UI, shared design system (e.g. `src/components/ui/`).
- TanStack Query, React Hook Form, Zod, Axios; Recharts for charts.

Long-term scope:

- **Multi-page apps per pillar** — Dedicated areas for Gaming, DeFi, NFT Marketplace, Launchpad, Governance (already started: Index, Gaming, DeFi, NFTMarketplace, Launchpad, Governance).
- **Wallet integration** — Connect Wallet (MetaMask, WalletConnect, etc.); account and chain switching; signing for transactions and (optional) meta-tx.
- **On-chain data** — Balances, allowances, pool state, NFT ownership; real-time or polling; optimistic updates where appropriate.
- **Off-chain data** — User profile, history, notifications; served by backend APIs.

- **Responsive & accessible** — Mobile-first where needed; a11y and i18n as the product grows.
- **Performance** — Code splitting per route, lazy loading, and caching to keep LCP and TTI acceptable as features grow.
- **State & resilience** — Clear handling of RPC/backend errors, pending txs, and network switches.

Why it takes time: Each pillar has many screens and states; wallet and chain handling is subtle; design system and UX must stay consistent across a large surface area.

3.2 Backend (Services & APIs)

Role: User identity, off-chain data, indexing, and orchestration; bridge between frontend and chains.

Current / baseline stack:

- Node.js, Express, MySQL, bcrypt, CORS, dotenv.
- REST APIs: `/api/customer`, `/api/agent`, `/api/policy`, `/api/statements`, `/api/centre`, `/api/contact`.

Long-term scope:

- **Auth & identity** — Signup, login, profile, session/JWT; optional link between wallet and email account; 2FA and recovery flows.
- **User and relationship data** — Customers, agents, policies, centre (link tables), statements; extended for gaming (achievements, inventory cache), DeFi (positions, history), marketplace (listings, offers), launchpad (participation, KYC state), governance (delegation, history).
- **Indexing & APIs** — Index on-chain events (transfers, pool changes, NFT listings, votes) for fast list/detail APIs; sync jobs and idempotency.
- **Notifications** — In-app and (optional) email/push for rewards, trades, listings, governance.
- **Admin & ops** — Internal APIs or admin UI for support, moderation, and launchpad vetting (with proper access control).
- **Security** — Rate limiting, input validation, SQL safety (parameterized queries), secret management; audit logs for sensitive actions.
- **Scalability** — Stateless API design; DB connection pooling, read replicas, and caching (e.g. Redis) as traffic grows.
- **Multi-chain coordination** — Which chain a user is on; storing per-chain state; optional relayer or backend-signed meta-tx.

Why it takes time: Data model grows with each pillar; indexing multiple chains is complex; auth and security must be done right from the start and refined over time.

3.3 Smart Contracts

Role: On-chain logic for token, rewards, marketplace, launchpad, and governance.

Current / baseline:

- ERC-20 (e.g. APOM) with trusted forwarder support.
- ERC-2771 Forwarder for meta-transactions (gasless / relayer-paid).
- RPC and ABI config for many chains (Ethereum, Polygon, BSC, Arbitrum, Avalanche, Fantom, Harmony, Heco, Klaytn, Moonbeam, Optimism, Palm, Ronin, xDai); each chain has middleware in the backend.

Long-term scope:

- **Token** — ERC-20 (and possibly governance wrapper); mint/burn roles; vesting and distribution contracts.
- **Meta-transactions** — ERC-2771 (and/or Biconomy, etc.) for better UX; nonce and replay protection; relayer infrastructure.
- **DeFi** — AMM (e.g. Uniswap V2/V3 style or custom), LP tokens, gauge/farm contracts, staking contracts; per-chain deployment and parameter tuning.
- **NFT Marketplace** — Listing, auction, offers; royalty enforcement; possibly shared or per-collection contracts.
- **Launchpad** — Sale contracts (fixed price, auction, etc.), vesting, refunds; timelock or multisig for admin.
- **Governance** — Governor + timelock (e.g. OpenZeppelin style); proposal creation, voting, execution.
- **Multi-chain** — Deploy and verify on each target chain; consistent interfaces where possible; chain-specific gas and config.
- **Upgrades & safety** — Proxy patterns where needed; pause roles; formal audits and bug bounties.

Why it takes time: Each contract family needs design, tests, internal review, external audit, and deployment/monitoring per chain. One mistake can mean loss of funds or reputation.

3.4 Infrastructure, Data & Operations

Role: Run the platform reliably and observe it.

Planned scope (over the year):

- **Environments** — Dev, staging, production; secrets and config per env.
- **Hosting** — Frontend (e.g. CDN + static hosting); backend (e.g. Node on VMs or containers); DB (managed MySQL or equivalent).
- **RPC** — Multiple providers per chain; failover and rate-limit handling; optional self-hosted nodes for critical paths.
- **Indexing** — Event indexing (self-built or The Graph, etc.); sync latency and backfill strategy.
- **Monitoring & alerting** — Health checks, error rates, latency, RPC and contract call failures; PagerDuty or similar.
- **Logging & tracing** — Structured logs; request IDs; optional distributed tracing.

- **Backup & DR** — DB backups, restore procedures, and incident runbooks.
- **Documentation** — Architecture, API, runbooks, and onboarding for new engineers.

Why it takes time: Setting up and hardening envs, monitoring, and runbooks is ongoing; multi-chain RPC and indexing alone are a continuous effort.

Part 4 — Multi-Chain & Smart Contract Strategy (Depth)

4.1 Supported Chains (Current Structure)

The backend is structured to support **many EVM chains** via dedicated middleware and shared contract caller:

- **Ethereum Mainnet** (e.g. Infura)
- **Polygon** (mainnet and alternate)
- **BSC** (Binance Smart Chain)
- **Arbitrum, Avalanche, Fantom, Harmony, Heco, Klaytn**
- **Matic, Moonbeam, Optimism, Palm, Ronin, xDai**

Each chain has:

- RPC URL(s) and, where used, contract addresses.
- ABI and error-handling shared where possible (e.g. `contractCaller.js`, `constant/constant.js`).

4.2 Contract Abstractions in Use

- **ERC-20** — `transfer`, `approve`, `allowance`, `balanceOf`, `decimals`, `name`, `symbol`, `totalSupply`; `trustedForwarder` and `isTrustedForwarder` for meta-tx.
- **ERC-2771 Forwarder** — `execute`, `verify`, `getNonce`, EIP-712 domain; used for gasless or relayer-paid user actions.

Future work will add: AMM pools, LP tokens, gauges, staking, NFT marketplace, launchpad sales, and governance (Governor + timelock).

4.3 Why Multi-Chain Is Hard

- **Consistency** — Same product behavior across chains despite different gas, block times, and finality.

- **Deployment** — Deploy and verify the same (or compatible) contracts on each chain; manage addresses and config.
- **Indexing** — Per-chain event indexing and unified APIs for the frontend.
- **Bridges & liquidity** — If users move assets between chains, bridge integration and liquidity considerations.
- **Support** — Debugging user issues requires chain, tx hash, and contract state; support playbooks and tools.

This is a major reason the project is planned over **1+ year**.

Part 5 — Security, Compliance & Risk

5.1 Security

- **Smart contracts** — Internal review, automated tests, external audits, and (optionally) bug bounties before and after mainnet.
- **Backend** — Auth (bcrypt, sessions/JWT), parameterized queries, rate limiting, and least-privilege access to DB and secrets.
- **Frontend** — No sensitive secrets in client; safe handling of wallet and signing; clear user confirmations for destructive or high-value actions.
- **Infrastructure** — Secured envs, secrets management, and minimal public surface.

5.2 Compliance & Legal

- **Launchpad** — KYC/AML where legally required; terms of use and investor eligibility; regional restrictions if applicable.
- **Token** — Clarity on utility vs security; legal review per jurisdiction for listing and distribution.
- **Data** — Privacy policy, data retention, and (where applicable) GDPR or similar.

Compliance work is ongoing and affects design and timelines.

5.3 Operational Risk

- **Smart contract risk** — Bugs, economic attacks, upgrade mistakes; mitigated by audits, testing, and phased rollout.
- **RPC and infra** — Downtime or rate limits; mitigated by multiple RPCs and monitoring.
- **Key management** — Admin and relayer keys; multisig and procedures to avoid single points of failure.

Part 6 — Phased Delivery (12–18 Month Horizon)

The following is an **illustrative** roadmap. Actual phases and dates depend on team size and priorities.

Phase	Focus	Example outcomes (by end of phase)
Phase 1 (Months 1–3)	Foundation & one chain	Auth and user APIs solid; ERC-20 + forwarder deployed on one mainnet; frontend connects wallet and shows balance; basic Gaming and DeFi landing and navigation.
Phase 2 (Months 4–6)	DeFi core & second chain	AMM + LP + staking on 1–2 chains; swap and add liquidity in UI; backend indexing for positions and history; security review of new contracts.
Phase 3 (Months 7–9)	Gaming & NFTs	At least one full game integration; NFT marketplace (list/buy/sell) on 1–2 chains; inventory and collection views; royalty handling.
Phase 4 (Months 10–12)	Launchpad & governance	Launchpad sale contracts and UI; KYC integration where needed; governance (proposals + voting); multi-chain expansion.
Phase 5 (Months 13–18)	Scale & polish	More chains and games; performance and UX improvements; audits and bounties; analytics, admin tools, and support readiness.

Ongoing throughout: monitoring, incident response, documentation, and dependency upgrades.

Part 7 — Team & Roles (What We Need to Build This)

To deliver this over 1+ year, we expect to rely on (or grow into) roles such as:

- **Frontend** — React/TypeScript engineers for web app, wallet integration, and multi-chain UX.
- **Backend** — Node/Express (or equivalent) engineers for APIs, indexing, auth, and integrations.
- **Smart contracts** — Solidity developers for token, DeFi, NFT, launchpad, and governance; familiarity with audits and multi-chain deploy.

- **DevOps / platform** — CI/CD, environments, RPC, monitoring, and on-call.
- **Product / design** — UX, flows, and prioritization across five pillars.
- **Security & compliance** — Reviews, audits coordination, and compliance checks.

Candidates may join for one primary area but will see the full picture and collaborate across frontend, backend, and contracts.

Part 8 — For Candidates: What You're Signing Up For

8.1 Scale of the Project

- **Five product pillars** — Gaming, DeFi, NFT Marketplace, Launchpad, Governance — each with substantial feature sets and multi-month workstreams.
- **Multi-chain** — Many EVM chains; contracts, RPC, and indexing per chain.
- **High bar** — Security, compliance, and reliability appropriate for handling real value and users.
- **Timeline** — We are planning for **at least 12 months** to reach a mature, production-ready platform, with continued iteration after that.

8.2 What You Might Work On

- **Frontend** — Building and refining flows for Gaming, DeFi, NFT Marketplace, Launchpad, and Governance; wallet and chain switching; performance and accessibility; design system and component library.
- **Backend** — Designing and implementing REST (or GraphQL) APIs; database schema and migrations; auth and sessions; indexing on-chain events; notifications and admin tools; multi-chain coordination.
- **Smart contracts** — Implementing and testing token, DeFi (AMM, gauges, staking), NFT marketplace, launchpad, and governance contracts; deploying and verifying on multiple chains; working with auditors and upgradeability.

8.3 Helpful Background

- **Frontend:** Strong React and TypeScript; state management and async data; REST/API integration; interest in Web3 (wallets, signing, RPC).
- **Backend:** Node.js and Express (or similar); SQL and schema design; auth and security basics; APIs and background jobs.
- **Smart contracts:** Solidity; EVM and common standards (ERC-20, ERC-721, ERC-1155); testing (e.g. Hardhat/Foundry); multi-chain and gas awareness.

Pluses: Vite, Tailwind, Radix; meta-transactions (ERC-2771); DeFi (AMM, staking); NFT marketplaces; gaming or launchpad experience; DevOps or monitoring.

Part 9 — Summary

APOM-DAPP is a **long-term, platform-level initiative** to build a **decentralized gaming and DeFi product** with five pillars: **Gaming Hub**, **DeFi Exchange**, **NFT Marketplace**, **Launchpad**, and **Governance**. It is designed for **multi-chain**, **security**, and **scalability**, and is expected to take **12–18+ months** to reach a mature, production-ready state, with ongoing iteration beyond that.

We are looking for people who want to own meaningful parts of this stack — **frontend**, **backend**, or **smart contracts** — and who are comfortable with the scope, complexity, and timeline of a serious Web3 platform.